

Bluestock Mutual Fund Analytics Capstone Project

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Project Overview

- Objective:
 - Analyze Indian Mutual Fund industry data and build analytics, risk models, and dashboard solutions.
- Datasets:
 - • Fund Master
 - • NAV History
 - • AUM
 - • SIP Inflows
 - • Investor Transactions
 - • Portfolio Holdings
 - • Benchmark Indices

Architecture

- Raw Data



- ETL Pipeline



- SQLite Database



- EDA



- Performance Analytics



- Power BI Dashboard



- Advanced Analytics

EDA Highlights

- nav_trend.png
- aum_growth.png
- sip_inflow.png

Key findings:

- Industry AUM grew strongly
- SIP inflows reached record levels
- Equity schemes dominated flows

Performance Analytics

Sharpe Ratio

Sortino Ratio

Alpha

Beta

Maximum Drawdown

Fund Scorecard

40 schemes analysed

Industry Overview

Industry Overview

Total AUM (₹ Lakh Cr)

12.50

Max of aum_lakh_crore

31K

Max of sip_inflow_crore

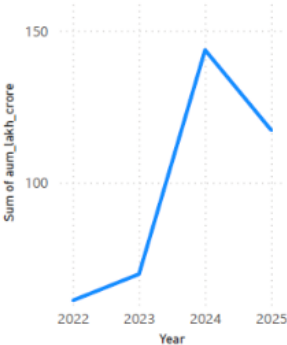
26.12

Max of total_folios_crore

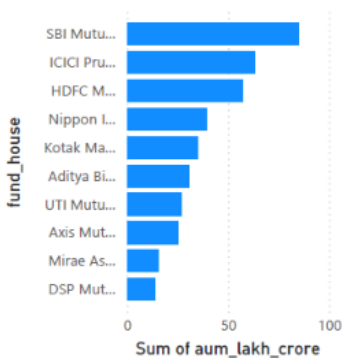
40

Count of amfi_code

Industry AUM Trend (2022–2025)



Sum of aum_lakh_crore by fund_house



Fund Performance Dashboard

Fund Performance

amfi_code	Sum of score	fund_house	scheme_name	category
100016	22.13	HDFC Mutual Fund	HDFC Top 100 Fund - Regular Plan - Growth	Equity
100025	22.13	HDFC Mutual Fund	HDFC Short Term Debt Fund - Regular - Growth	Debt
100033	77.25	HDFC Mutual Fund	HDFC Mid-Cap Opportunities Fund - Regular - Growth	Equity
101206	57.19	Aditya Birla Sun Life MF	ABSL Frontline Equity Fund - Regular - Growth	Equity
101207	41.81	Aditya Birla Sun Life MF	ABSL Small Cap Fund - Regular - Growth	Equity
101208	21.88	Aditya Birla Sun Life MF	ABSL Liquid Fund - Regular - Growth	Debt
102885	49.75	UTI Mutual Fund	UTI Nifty 50 Index Fund - Regular - Growth	Equity
Total	2,050.00			

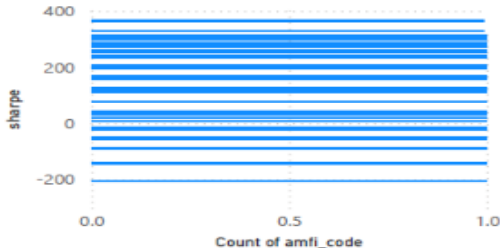
fund_house

All

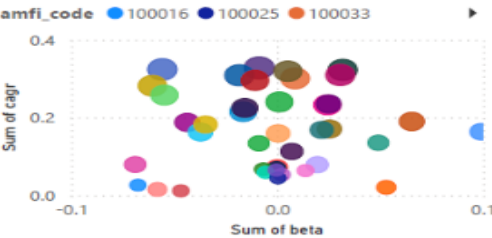
category

All

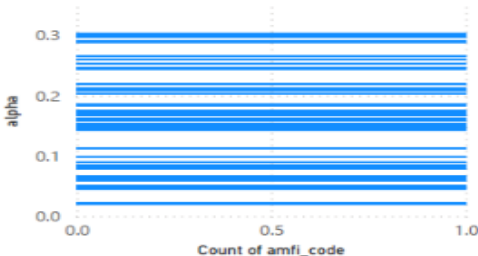
Top Funds by Sharpe Ratio



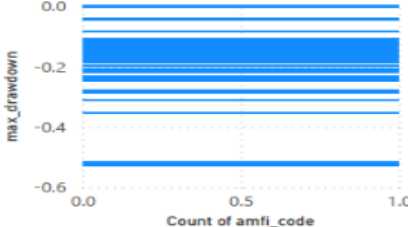
Risk vs Return Analysis



Alpha Ranking



Maximum Drawdown Comparison



risk_category

All

Power & Battery

Energy saver is on

SIP & Market Trends Dashboard

SIP & Market Trends

category

- ☐ ELSS
- ☐ Flexi Cap
- ☐ Gilt
- ☐ Hybrid

Sum of close_value

Year

month

4/1/2024

3/1/2025

Net Inflow by Category

net_inflow_crore

Monthly SIP Inflows

Sum of sip_inflow_crore

Year

Net Inflow by Category

net_inflow_crore

Category Inflow Heatmap

category	2024	2025	Total
ELSS	4627	1453	6080
Flexi Cap	47551	16438	63989
Gilt	7753	2642	10395
Hybrid	29711	9157	38868
Large & Mid Cap	43169	14583	57752
Large Cap	19449	6184	25633
Liquid	346328	104947	451275
Mid Cap	41116	14196	55312
Sectoral/Thematic	78107	25722	103829
Short Duration	41859	13671	55530
Total	706618	225621	932239

investor Analytics Dashboard

Investor Analytics

state

- ☐ Delhi
- ☐ Gujarat
- ☐ Haryana
- ☐ Karnataka
- ☐ Madhya Pradesh
- ☐ Maharashtra

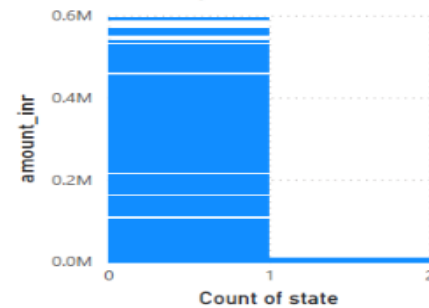
age_group

- ☐ 18-25
- ☐ 26-35
- ☐ 36-45
- ☐ 46-55
- ☐ 56+

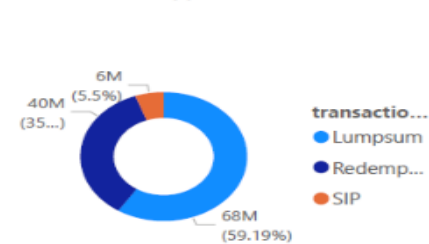
city_tier

- ☐ B30
- ☐ T30

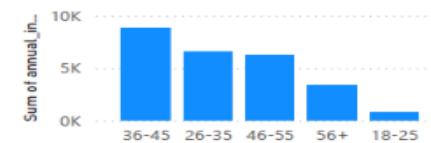
Count of state by amount_inr



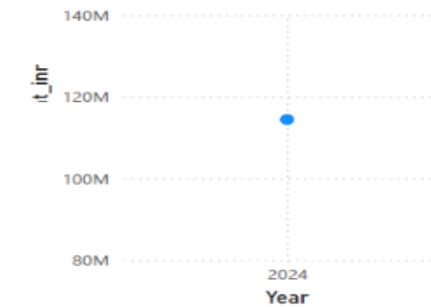
Transaction Type Distribution



Average Investment by Age Group

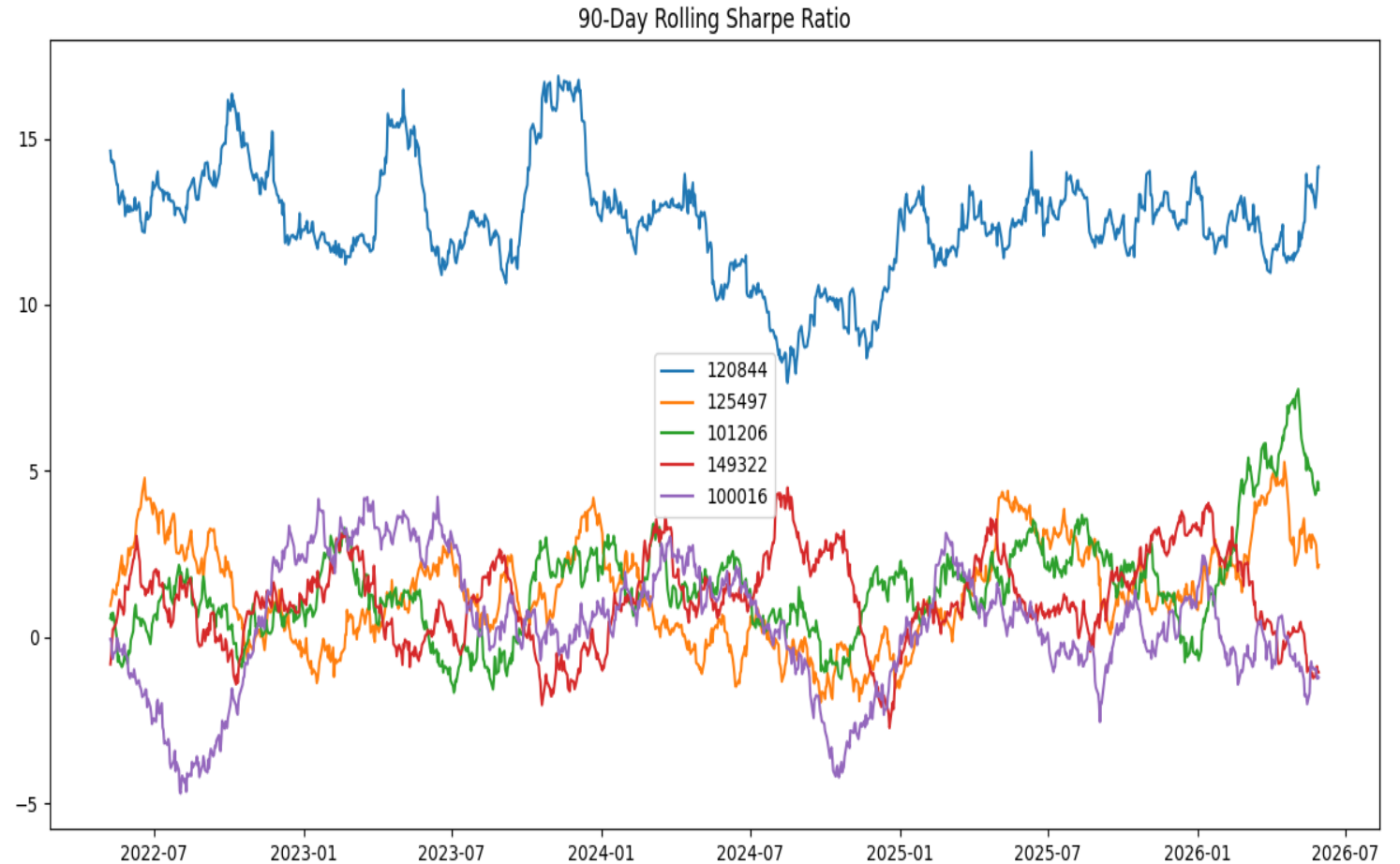


Monthly Transaction Volume



Advanced Analytics

- Historical VaR
- CVaR
- Rolling Sharpe
- Cohort Analysis
- SIP Continuity
- Fund Recommender
- HHI Concentration



Key Insights

- 1. SBI MF led industry AUM growth.
- 2. SIP inflows crossed ₹31,000 Cr monthly.
- 3. Risk-adjusted performance varies significantly.
- 4. SIP continuity remained strong.
- 5. Portfolio concentration differs across funds.

Conclusion

- Built end-to-end mutual fund analytics platform
- covering ETL, data engineering, analytics,
- risk modeling and business intelligence.